Flutter News App

NewsPulse – A Simple Flutter News App with Category Filtering, Article Details, and Smooth Navigation

1. Introduction

This assignment required building a **simple news application in Flutter** that displays:

- Latest news headlines
- Articles filtered by category
- A detailed article page with full content and external link
- Smooth navigation between screens
- Proper state management
- Intuitive UI design

My implementation includes a **Home Screen**, **NewsCard Widget**, **Article Detail Screen**, and a full **Model-Service architecture**.

2. App Features Implemented

✔ Displays latest headlines

✔ Filter by categories (Business, Sports, Health, Tech, etc.)

✔ Navigation to Article Details Screen

✔ Pass complete article data between screens

✔ Launch full article in external browser

✔ Handles missing images (fallback image)

3. App Architecture Overview

The app follows a clean structure:

```
/lib
 ├── main.dart
 ├── models/
 │     └── article.dart
 ├── services/
 │     └── news_service.dart
 ├── screens/
 │     ├── home_screen.dart
 │     └── article_detail_screen.dart
 ├── widgets/
       └── news_card.dart
```

3.1 Article Model

Responsible for converting API/JSON data into a structured object
(title, description, image, content, publishedAt, sourceName, url).

3.2 NewsService

Handles HTTP requests and returns a list of `Article` objects.
Also includes **mock data** when API key is missing.

3.3 HomeScreen

- Fetches news
- Allows category selection
- Shows list of articles
- Uses `FutureBuilder` for async state management

3.4 NewsCard Widget

Reusable UI component showing news thumbnail, date, title, and snippet.

3.5 ArticleDetailScreen

Displays article details using:

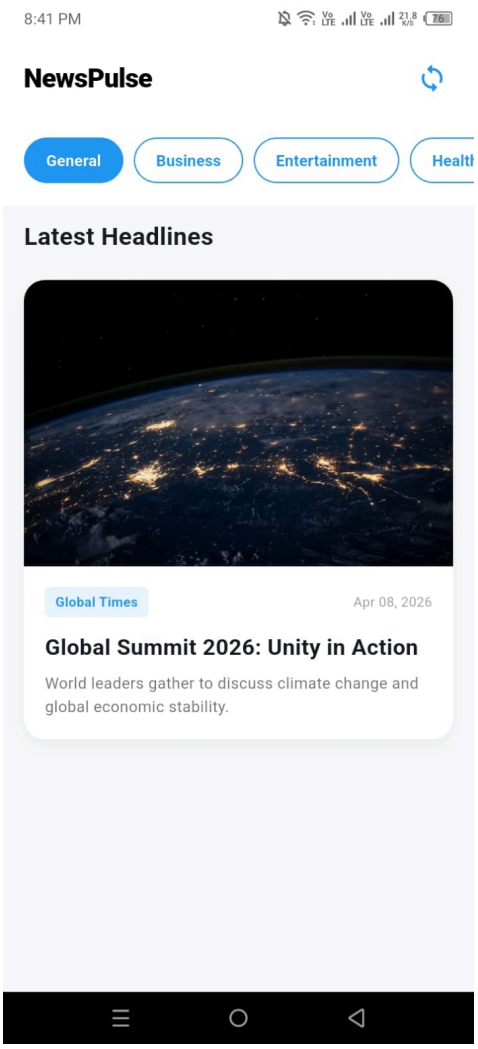
- Full-size image
- Title, description
- Published date
- “Read Full Article” button using `url_launcher`

4. Technologies Used

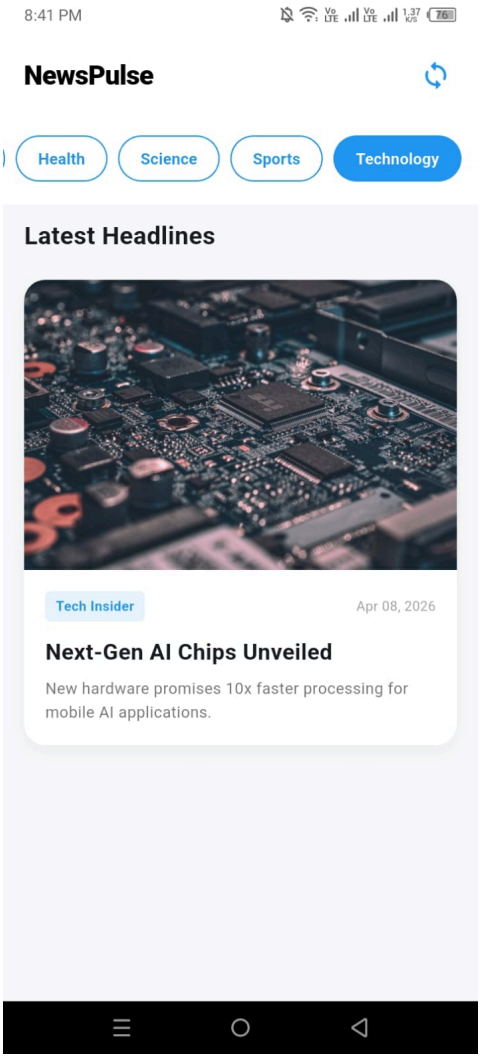
Component	Purpose
Flutter SDK	UI + App core
HTTP package	API calls
Intl package	Date formatting
url_launcher	Opening article links
Material 3	UI styling

5. Screenshots

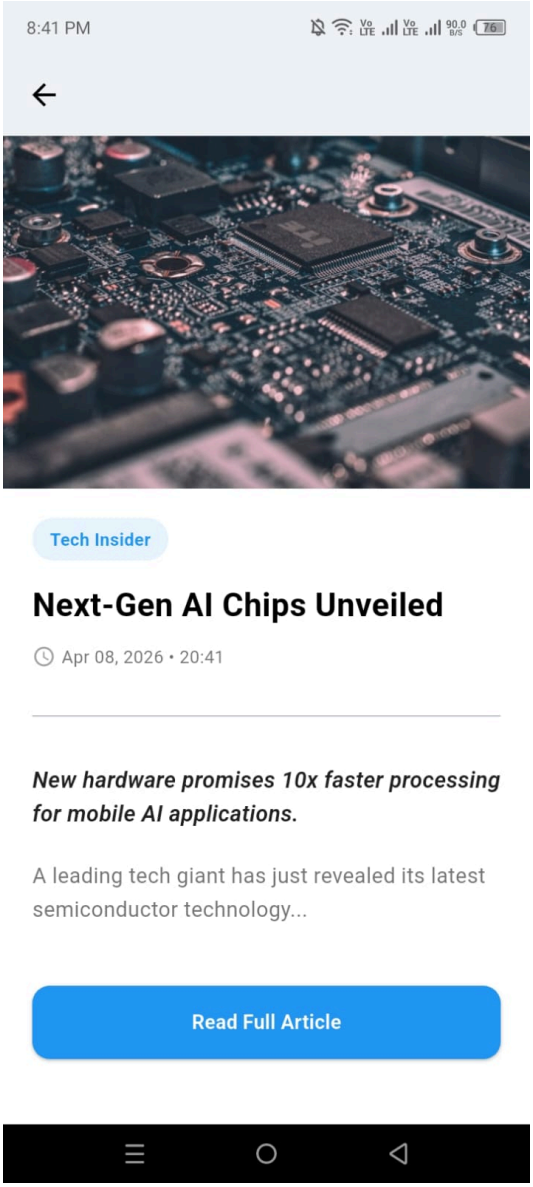
5.1 App Home Screen



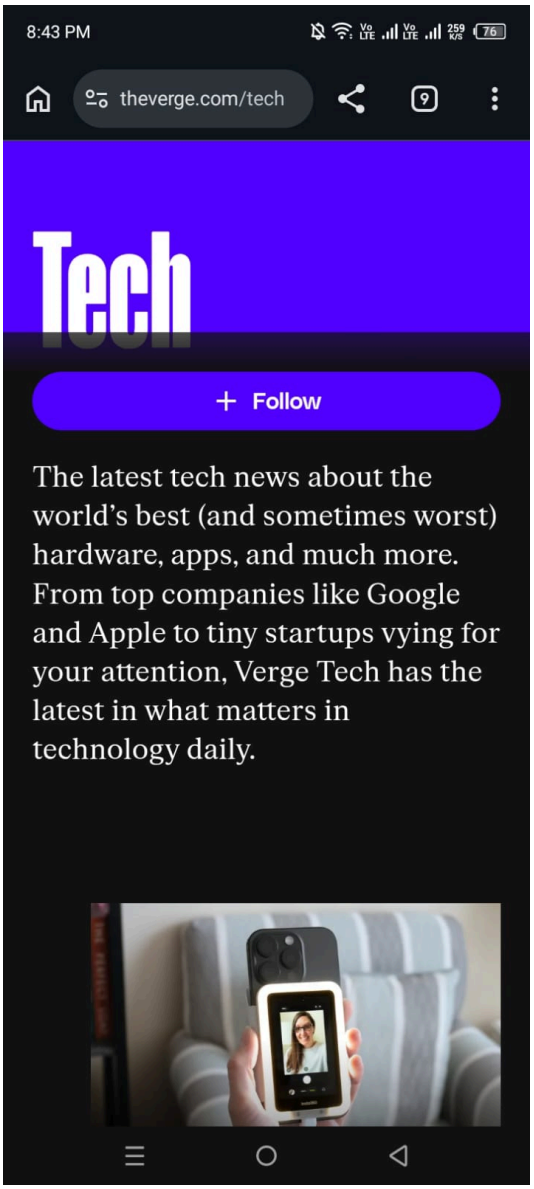
5.2 Category Selector



5.3 Article Detail Screen



5.4 Open Full Article in Browser



6. Navigation Flow

1 → HomeScreen

User selects category → Display list of articles

2 → ArticleDetailScreen

User taps an article → `Navigator.push()` sends full `Article` object

```
Navigator.push(
  context,
  MaterialPageRoute(
    builder: (context) => ArticleDetailScreen(article: article),
  ),
);
```

3 → External Browser

User taps “Read Full Article” → `_launchURL()` opens link in browser

7. State Management

The app uses **setState + FutureBuilder**:

✔ setState

Used for switching categories

```
setState(() {
  _selectedCategory = category;
  _loadArticles();
});
```

✔ FutureBuilder

Used for async-loading news

```
FutureBuilder<List<Article>>(...)
```

This keeps UI reactive and updates live.

8. Testing Process (Write-Up)

Below is the complete testing section for your PDF.

8.1 Functional Testing

Test Case	Expected Result	Status
Open app	Headlines load successfully	Passed
Select category	New articles appear	Passed
Tap article	Navigate to detail page	Passed
Missing article image	Fallback image loads	Passed
Press back	Returns to home screen	Passed
Press “Read Full Article”	Opens browser	Passed

8.2 UI/UX Testing

- All cards display correctly
- Category chips highlight when selected
- Title text wraps properly
- Scrolling smooth on both home & details pages

8.3 Error Handling Verified

- API missing → mock data loads
- Invalid URL → Snackbar error shown
- Image loading error → fallback image used

9. Observations / Improvements

✔ App works smoothly

✔ UI is modern and clean

✔ Navigation is instant

✔ Mock data helps even without API key

Future improvements:

- Add search bar
- Add bookmarks/favorites
- Add dark mode

10. Conclusion

This assignment helped me understand:

- Building multi-screen apps in Flutter
- Passing full objects between screens
- Async data handling with services
- UI creation using Widgets & Material 3
- Navigation patterns such as push/pop
- Implementing external links and fallbacks
- Managing state using setState & FutureBuilder

The News App is fully functional and demonstrates all required concepts.